

EXERCISE SHEET 11

GROUP AND RING THEORY

11.1. Recall that we say that a ring R is *simple* if the only ideals of R are (0) and R itself. For a division ring D , show that the ring $M_n(D)$ of $n \times n$ matrices over D is a simple ring.

11.2. A ring R is a *prime* ring if (0) is a prime ideal in R .

Let R be a prime left Artinian ring with unity. Show that R is isomorphic to an $n \times n$ matrix ring over a division ring. Hence, show that a prime ideal in an Artinian ring with unity is maximal.

EXTRA QUESTIONS

11.3. For a field F , consider the quotient ring $R = F[x_1, x_2, \dots]/(\{x_n^n \mid n \in \mathbb{N}\})$. Find nilpotent ideals $A_1, A_2, \dots$ such that the sum $\sum_i A_i$ is not a nilpotent ideal.

11.4. For a ring R and $a \in R$, show that the (2-sided) ideal generated by a is given by

$$(a) = aR + Ra + RaR + a\mathbb{Z}$$

where $RaR = \{\sum_{i=1}^{\ell} r_i a s_i \mid \ell \in \mathbb{N}, r_i, s_i \in R\}$.

11.5. Let R be a ring with unity. Let $\text{Hom}_R(R, R)$ denote the ring of endomorphisms of R regarded as a left R -module. Show that $R^{\text{op}} \cong \text{Hom}_R(R, R)$ as rings. [Recall that the opposite R^{op} of a ring R is another ring with the same elements and addition operation, but with the multiplication performed in the reverse order. That is, the opposite of a ring $(R, +, \cdot)$ is the ring $(R, +, *)$ whose multiplication $*$ is defined by $a * b = b \cdot a$ for all $a, b \in R$.]